Lecture 19: Calculus with Parametric Curves

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Differentiate parametric equations to find the first derivative and second derivative.
- ☐ Determine the slope of the tangent line at a given point.
- ☐ Find the area under a parametric curve when bounded by the x-axis or y-axis.
- ☐ Determine the length of a curve defined parametrically.

Derivative of a Parametrized Curve

A parametrized curve $x = f(t)$ and $y = g(t)$ is differentiable at t if f and g are each differentiable at t.

If y is differentiable function of x, then by the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If all three derivatives exist and $dx/dt \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

If y is twice-differentiable function of x, then at any point where $dx/dt \neq 0$ and $y' = dy/dx$,

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}.$$

Derivative of a Parametrized Curve

Example: Suppose that a curve $\mathcal{C}$ is defined as

$$x = t^2 - 1, \quad y = \frac{3}{t}.$$

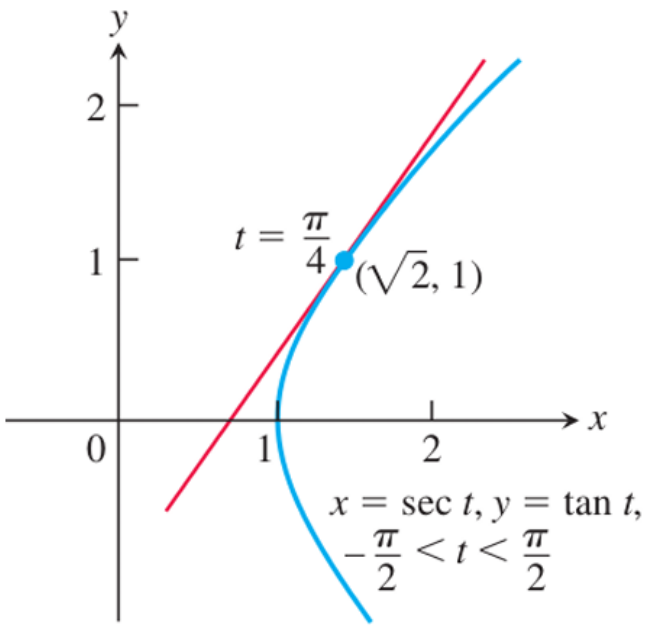
(a) Find a Cartesian relation between x and y.
(b) Which point on the curve corresponds to $t = 6$?
(c) Find $\frac{dy}{dx}$ at the point $(8, 1)$ by using the parametric differentiation.

Tangent Lines and Areas

Example: Find the tangent line to the curve

$$x = \sec t, \quad y = \tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2},$$

at the point $(\sqrt{2}, 1)$, where $t = \frac{\pi}{4}$.



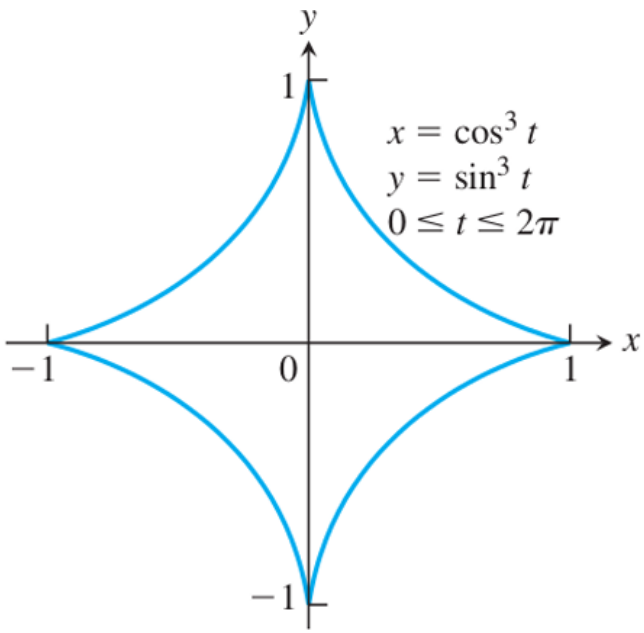
Tangent Lines and Areas

Example: Find $\frac{d^2y}{dx^2}$ as a function of t if $x = t - t^2$ and $y = t - t^3$.

Tangent Lines and Areas

Example: Find the area enclosed by the astroid

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \leq t \leq 2\pi.$$



Length of a Parametrically Defined Curve

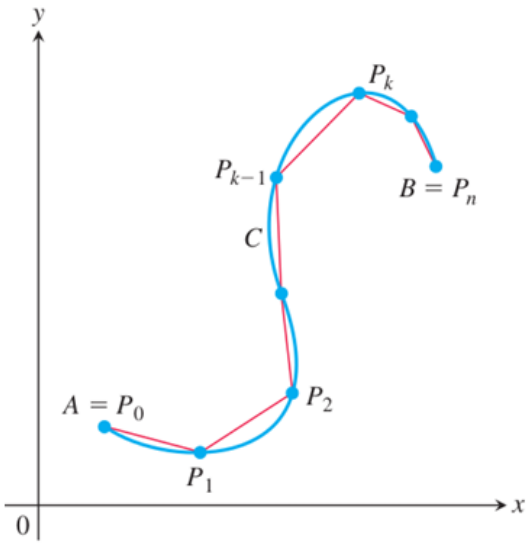
A curve $\mathcal{C}$, given by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, is a smooth curve if f and g are continuously differentiable on the interval $[a, b]$ with f' and g' are not simultaneously zero.

Length of a smooth curve

Let $\mathcal{C}$ be a smooth curve given by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$. If $\mathcal{C}$ is traversed exactly once at t increases from $t = a$ to $t = b$, then the length of $\mathcal{C}$ is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\ dt.$$

Length of a Parametrically Defined Curve



Length of a Parametrically Defined Curve

Example: Find the length of the circle of radius r defined parametrically by

$$x = r\cos t \quad \text{and} \quad y = r\sin t, \quad 0 \leq t \leq 2\pi.$$

Length of a Parametrically Defined Curve

Example: Find the length of the astroid

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \leq t \leq 2\pi.$$